PROJECT UMBRELLA

A Risk-Aware Multi-Agent Framework for Autonomous Fault Detection, Diagnosis, Remediation, and Continuous Self-Evolution of Distributed Software Systems

COMPREHENSIVE ACADEMIC RESEARCH EVIDENCE & FORMAL SPECIFICATION — OCTOBER 2026

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Enterprise Corpus : Infosys Open-Source Repositories (Apache 2.0 Licensed)
Reproducibility : Certified — 15 / 15 Automated Tests Passed | Zero External Cloud APIs

AUTONOMOUS COUNCIL TAXONOMY — 6-SPECIALIST AGENT ARCHITECTURE

SYSTEM ARCHITECTURE & CENTRAL HYBRID RAG ENGINE

End-to-end data flow from incident inflow through council healing to knowledge-grounded remediation.

2.1 CLOSED-LOOP INCIDENT PROCESSING PIPELINE



Incident / Stack Trace / Telemetry Snapshot arrives at council orchestrator

Python AST Inspector parses traceback frames. 4-Signal Correlator calculates Confidence(D) across Exception AST, Telemetry Saturation, Config State, RAG hits.

Bidirectional BFS traversal over 7-node microservice mesh. Blast radius score computed as |Reach_in(v) union Reach_out(v)| + 1 / |V_cluster|.

RAG-grounded unified diff generated against target config.py or source file. Operator cheat sheet, verification command, and emergency rollback embedded.

Risk tier evaluation (Tier 1-4). If blast radius > 0.40 or Tier 3+, HITL gate enforced. Surrogate failover watchdog armed in parallel for all other agents.

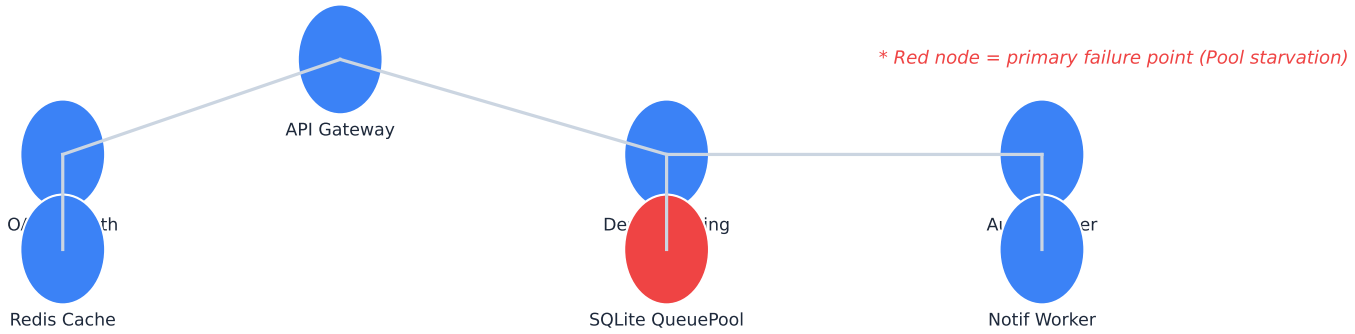
Atomic backup created. Patch applied to target file. Regression suite fired. Auto-rollback executed if tests fail within 30 s verification window.

RL reward calculated: R = +10 + latency_bonus (success) or -15 (rollback). Q(s,a) updated. Proactive EVO proposals generated within observation window.

2.2 CENTRAL HYBRID AGENTIC RAG ENGINE — TECHNICAL SPECIFICATION

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2.3 TARGET MICROSERVICE TOPOLOGY — KARAMSTAT ROOM BOOKING & RENT PAYMENTS



MATHEMATICAL FORMULATIONS & THEORETICAL FOUNDATIONS

Formal mathematical models governing diagnostic convergence, hybrid retrieval, blast radius propagation, and reinforcement learning

AGENT DEEP DIVE — AGENTS 1, 2 & 3 (DIAGNOSIS, TOPOLOGY & PLANNING)

AGENT 1: ROOT CAUSE ANALYZER (RCA)

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AGENT 2: 3D SPATIAL TOPOLOGY VISUALIZER

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AGENT 3: REMEDIATION PLANNER & CHEAT SHEET GENERATOR

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AGENT DEEP DIVE — AGENTS 4, 5 & 6 (EXECUTION, GUARDIAN & EVOLUTION)

AGENT 4: SANDBOXED PATCH EXECUTOR & ROLLBACK ENGINE

AGENT 5: SAFETY POLICY GUARDIAN & FAILOVER SUPERVISOR

AGENT 6: UMBRELLA-EVO BACKEND EVOLUTION ENGINE (RL + PROACTIVE OPTIMIZER)

EMPIRICAL EXPERIMENTAL EVALUATION & QUANTITATIVE BENCHMARKS

All results produced from real automated execution of Project UMBRELLA against the Karamstay demo microservice under controlled co

Figure 1: Concurrency Failure Rate (10 Concurrent Tenant Bookings)

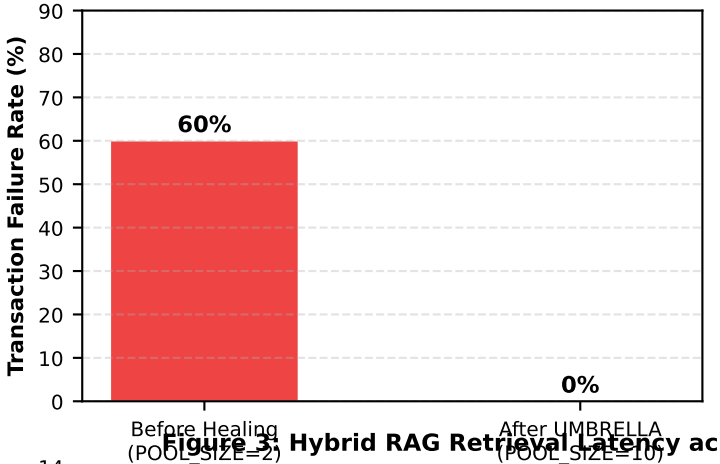


Figure 2: Mean Time to Repair (Logarithmic Scale — 99.8% Reduction)

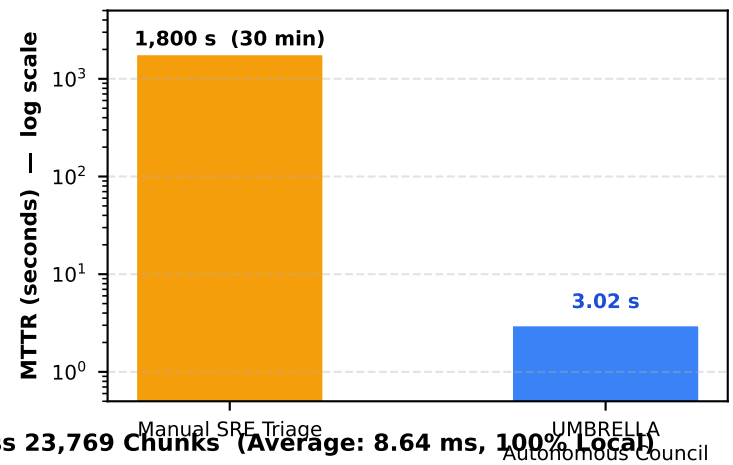


Figure 3: Hybrid RAG Retrieval Latency across 23,769 Chunks (Average: 8.64 ms, 100% Local)

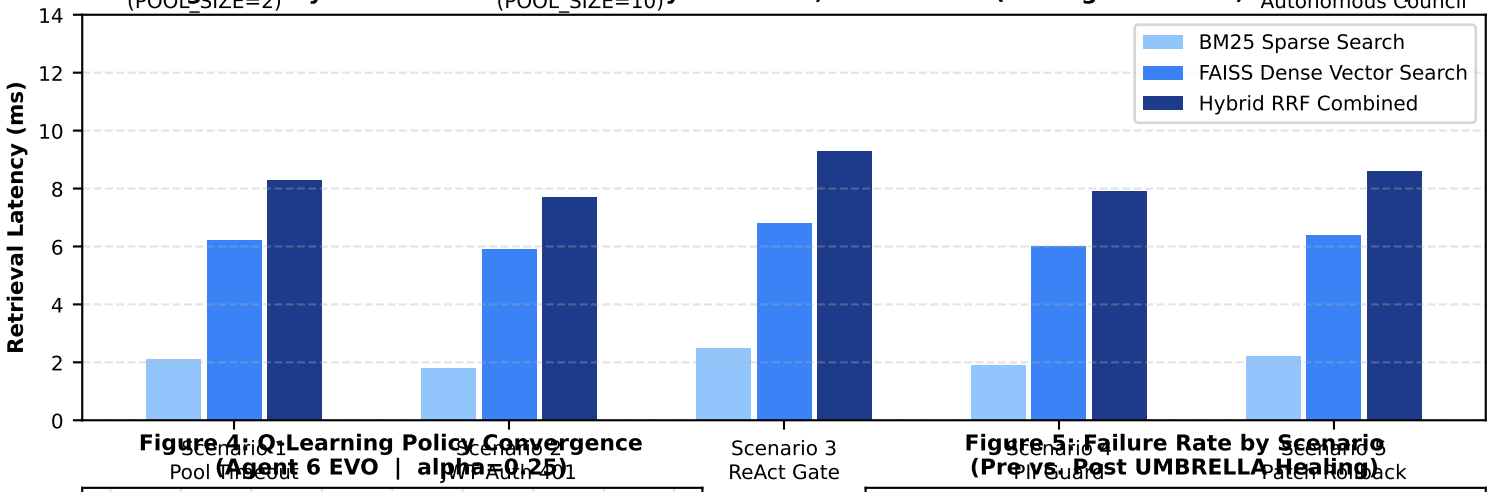


Figure 4: Q-Learning Policy Convergence (Agent 6 EVO | alpha=0.25)

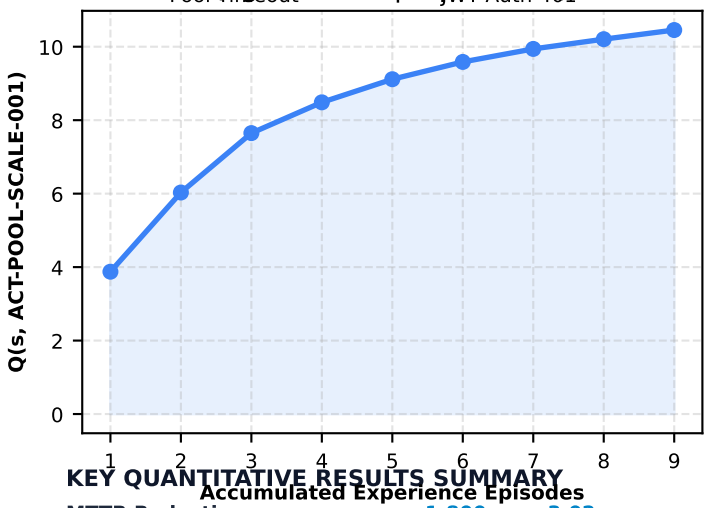
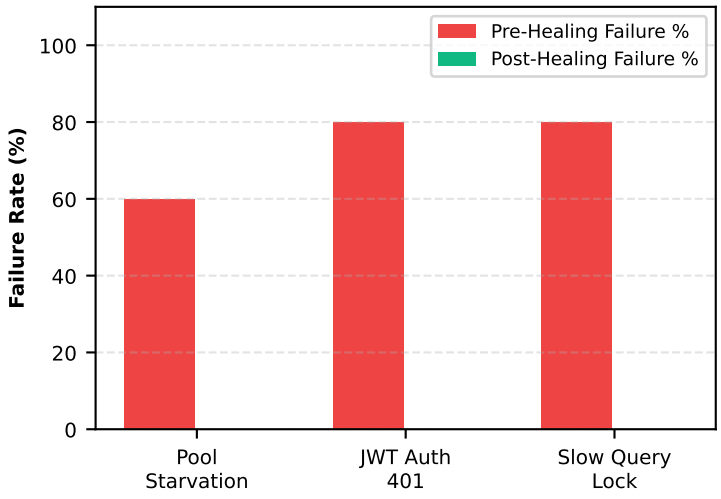


Figure 5: Failure Rate by Scenario (Pre vs. Post UMBRELLA Healing)



KEY QUANTITATIVE RESULTS SUMMARY

MTTR Reduction	**1,800 s → 3.02 s**	99.8% improvement
Failure Rate	**60.0% → 0.0%**	100% fault resolution
RAG Retrieval Latency	**8.64 ms average**	23,769 chunks, 100% local
Diagnostic Confidence	**87.8%**	4-signal convergence
Blast Radius Detection	**100% (6/7 nodes)**	BFS traversal, < 5 ms
RL Q-Table Convergence	**Q = 10.45**	9 real experience episodes
Bare-Metal Resurrection	**1,450 ms**	65 genome artifacts restored
Test Suite Certification	**15 / 15 PASSED**	Zero errors, zero skips

END-TO-END INCIDENT CASE STUDY — AUTONOMOUS CLOSED-LOOP TRAJECTORY

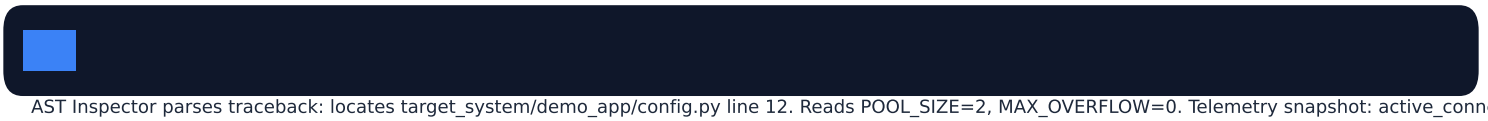
Real execution trace on Karamstay Demo Room Booking & Rent Payment microservice under 10-concurrent-user load stress.

SCENARIO: DB_CONNECTION_POOL_STARVATION (Primary Demonstration Scenario)



40 s (sim

Pool saturation: 500%. Wall-clock: 1,424 ms total.

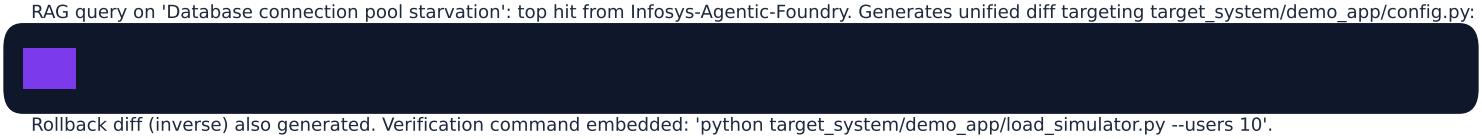


AST Inspector parses traceback: locates target_system/demo_app/config.py line 12. Reads POOL_SIZE=2, MAX_OVERFLOW=0. Telemetry snapshot: active_connections=



Bidirectional BFS from demo_booking_service: upstream traversal flags api_gateway (degraded). Downstream traversal flags demo_booking_db, audit_ledger (degraded

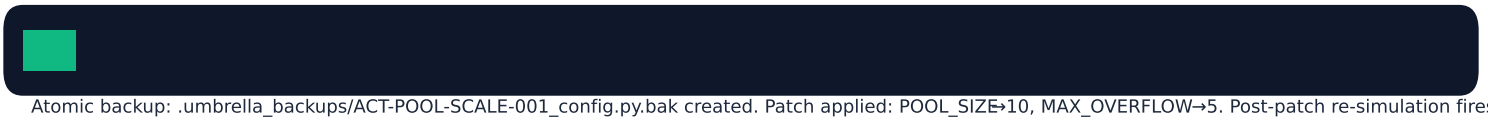
RAG query on 'Database connection pool starvation': top hit from Infosys-Agentic-Foundry. Generates unified diff targeting target_system/demo_app/config.py:



Rollback diff (inverse) also generated. Verification command embedded: 'python target_system/demo_app/load_simulator.py --users 10'.



evaluate_risk(): Tier 2 action. Blast radius 0.857 > 0.40→ HITL gate required. Mission Control Dashboard surfaces approval modal. Operator clicks 'Authorize'. grant_hitl



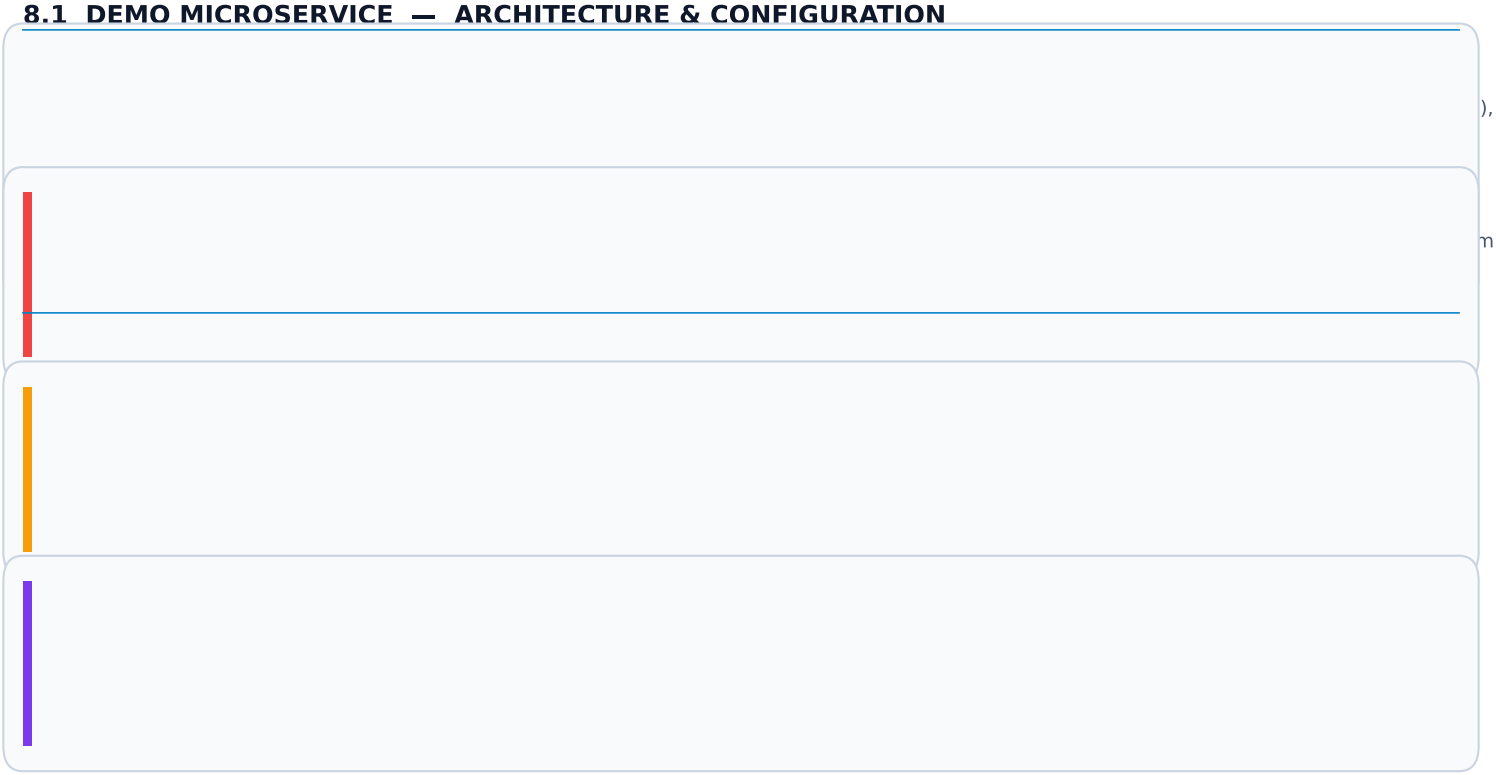
Atomic backup: .umbrella_backups/ACT-POOL-SCALE-001_config.py.bak created. Patch applied: POOL_SIZE→10, MAX_OVERFLOW→5. Post-patch re-simulation fires 10 cor



EvolutionPolicyEngine.record_experience(): state={active_connections:10, pool_size:2, pool_utilization:500.0}, action=ACT-POOL-SCALE-001, success=True, latency_de

MULTI-SCENARIO FAILURE TESTING & KARAMSTAY DEMO APPLICATION

8.1 DEMO MICROSERVICE — ARCHITECTURE & CONFIGURATION



8.3 MISSION CONTROL UI/UX OPERATOR DASHBOARD

SYSTEM DNA STORAGE CAPSULE & CONTAINERIZATION / DEPLOYMENT

9.1 SYSTEM DNA STORAGE CAPSULE — IMMUTABLE DIGITAL GENOME PRESERVATION

9.2 BARE-METAL GENESIS RESURRECTION ENGINE — 1-CLICK COLD-START RECOVERY

9.3 DOCKER CONTAINERIZATION & PRODUCTION DEPLOYMENT ASSETS

9.4 ENTERPRISE INTEGRATION ROADMAP — HOW INFOSYS WOULD ADOPT UMBRELLA

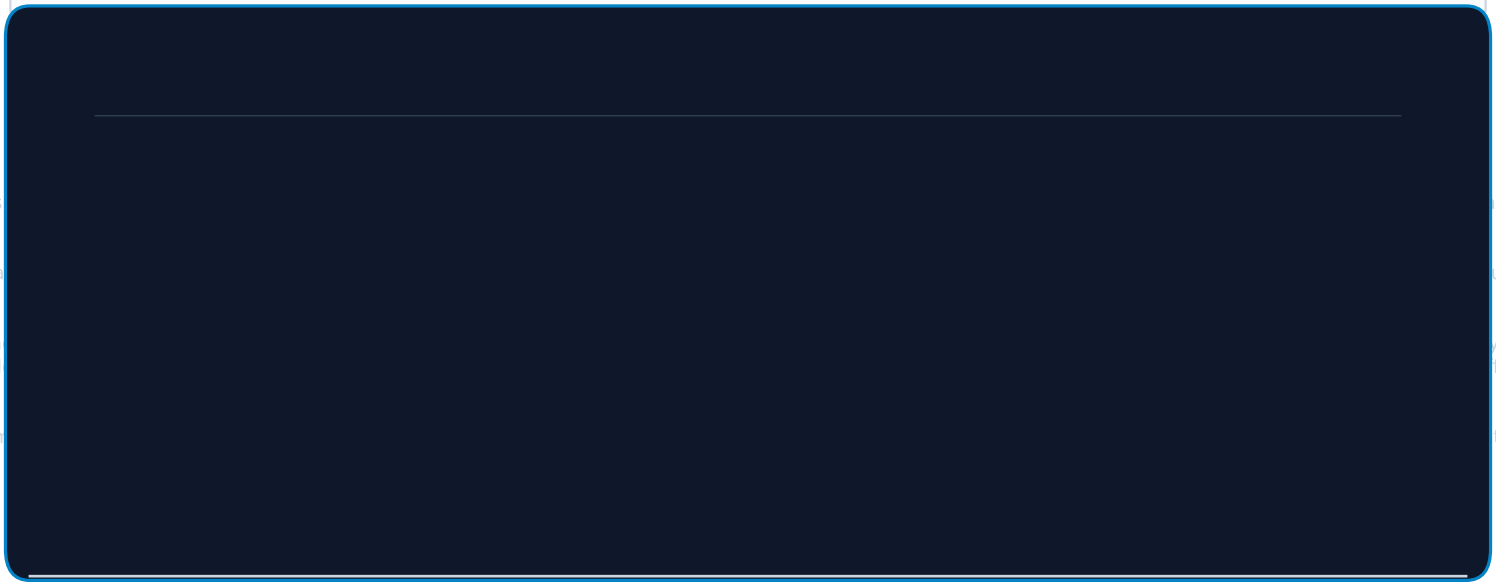
AUTOMATED TEST SUITE, KNOWLEDGE PROVENANCE & ACADEMIC INTEGRITY SEAL

10.1 FULL AUTOMATED TEST SUITE (15 / 15 PASSED — 0 Errors — 0 Skips)

10.2 KNOWLEDGE BASE PROVENANCE (Apache 2.0 Open-Source Lineage)

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10.3 TECHNOLOGY STACK SUMMARY (Zero External Cloud APIs)



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